

Obligatorisk oppgave 2 – IN2100

Oppgave 63 (1-4) – Use LPO to prove termination of the following specifications:

1. $\{f(f(x)) = f(x)\}$
2. $\{f(x) = g(x), g(b) = f(a)\}$
3. $\{f(g(x)) = g(f(x))\}$
4. $\{g(f(x)) = f(g(x))\}$

1.1 $f(f(x)) \succ_{lpo} f(x)$ follows from lpo-1, since f is the main function symbol in both places, if we can prove $f(x) \succ_{lpo} x$, which follows from lpo-1.

Since each rewrite step strictly decreases according to LPO, the system terminates.

1.2 $f(x) \succ_{lpo} g(x), f(a) \succ_{lpo} g(b)$. We assume $f > g$. On the first rule we have $f(x)$ on the left-hand and $g(x)$ on the right-hand. Following our assumption of $f > g$, this rewrite decreases the term according to LPO. On the second rule we have $g(b)$ on the left-hand and $f(a)$ on the right-hand. Since we assumed $f > g$, this rule increases the function precedence. Since the second rule increases under LPO, the rewrite system does not terminate. The terms can keep cycling as follows:

$$g(b) \rightarrow f(a) - g(a) \rightarrow f(a) \rightarrow g(a) \rightarrow \dots$$

1.3 We assume $f > g$. We have f as the left-hand outer function, and g as the right-hand outer function. Since $f > g$, it's guaranteed that it will decrease and therefore the system terminates.

1.4 We assume $f > g$. We have g as the left-hand outer function, and f as the right-hand outer function. Since $f > g$, this does not immediately guarantee decrease, so we must check recursively. The argument of the left-hand side is $f(x)$ and the argument of the right-hand side is $g(x)$. Since $f > g$, we know that $f(x) > g(x)$. And since LPO allows a function g to be greater than f if at least one of its arguments is greater, we conclude

$g(f(x)) \succ_{LPO} f(g(x))$. Which means this rewrite step is always decreasing under LPO, and therefore the system terminates.

Oppgave 71 (1-2) – Consider the specification $\{f(g(x)) = h(x), h(x) = g(f(x)), a = b\}$

1. Can the system be proved to terminate using weight functions?
2. Can the system be proved to terminate using lpo or mpo?

1.1 Using weight functions:

We first define a weight function w such that:

$$w(f(x)) > w(g(x))$$

$$w(f(x)) > w(h(x))$$

$$w(h(x)) > w(g(f(x)))$$

We check now the first rule $f(g(x)) \rightarrow h(x)$

We need $w(f(g(x))) > w(h(x))$

If we assume a simple additive weight function, such as:

$$w(f(t)) = w(t) + C_f, \quad w(g(t)) = w(t) + C_g, \quad w(h(t)) = w(t) + C_h$$

For termination, we need: $w(f(g(x))) > w(h(x))$

Which simplifies to: $w(x) + C_g + C_f > w(x) + C_h$

Canceling $w(x)$ from both sides gives us: $C_g + C_f > C_h$

Which can be satisfied if we choose $C_f > C_h - C_g$

On the second rule $h(x) \rightarrow g(f(x))$

We need $w(h(x)) > w(g(f(x)))$

Using our weight function:

$$w(h(x)) = w(x) + C_h, \quad w(g(f(x))) = w(f(x)) + C_g = (w(x) + C_f) + C_g$$

For termination, we need: $w(x) + C_h > w(x) + C_f + C_g$

Canceling $w(x)$ from both sides gives us: $C_h > C_f + C_g$

And for this to hold we need C_h to be strictly greater than $C_f + C_g$, which is possible.

Since we can assign weight values C_f, C_g, C_h such that all rewrites strictly decrease weight, the system is proven to terminate using weight functions.

2 Using LPO or MPO

We can't neither prove with LPO or MPO that the system terminates. This is due to coming to points where it's not guaranteed a strict decrease.

Under LPO we start defining a function precedence $f > h > g$. The first rule works under LPO since the outer function changes from f to h , and since $f > h$, this step decreases. On the

second rule, the outer function changes from h to g , which would be fine but the argument inside g is $f(x)$ which is larger than $h(x)$. Since $f(x) > h(x)$ we are not guaranteed a strict decrease under LPO. The rules do not form an infinite loop, so the system might still terminate but we can't prove it with LPO.

Under MPO we assume function precedence $f > h > g$. The first rule works since the left-hand argument is $g(x)$ and the right-hand argument is x . Since $g(x) > x$, this step is valid under MPO. On the second rule, the left-hand argument is x and the right-hand argument is $f(x)$. Since $f > h > g$, we normally have $f(x) > x$, meaning this step does not strictly decrease under MPO.

Oppgave 82 (1) – From [6]

1. Find terms r_1 and r_2 such that $\{f(g(x)) = r_1, g(h(x)) = r_2\}$ is confluent (and terminating).

Let $r_1 = h(x)$ and $r_2 = x$, then we have $f(g(x)) = h(x)$ and $g(h(x)) = x$.

Checking termination: Rule 1 decreases complexity because the term depth is reduced, and rule 2 further simplifies terms. Since there is no infinite loop, the system terminates.

Checking confluence: If we start with $g(h(x))$, we get x . If we substitute $g(x)$ inside $f(g(x))$, we get $h(x)$. Since neither rule introduces new conflicts or divergences, the system is confluent.